Dynamic Scene's Laser Localization by NeuroIV-Based Moving Objects Detection and LiDAR Points Evaluation

Xiangyong Liu, *Member, IEEE*, Zhixin Yang, *Member, IEEE*, Jing Hou, *Member, IEEE*, and Wei Huang

***Abstract*— Accurate localization is an important component of the vehicle's autonomous navigation. The appearance of the moving objects may lead to a feature-matching error with the map features, thereby causing a serious decline in localization accuracy. A neuromorphic vision (NeuroIV) sensor is a kind of dynamic vision sensor with the properties of high temporal resolution, movement capture, and lightweight computation. In view of this, this research proposes to combine the NeuroIV and LiDAR points to acquire the static landmark features and robust navigation localization. However, as a younger and smaller research field compared to RGB computer vision, NeuroIV vision is rarely associated with the intelligent vehicle. For this purpose, we built a novel dataset recorded by NeuroIV sensor, and a state-of-the-art YOLO-small network is designed to detect the moving objects with the dataset. In order to completely deduct the whole dynamic zones, a sensors' novel fusion model is built by the zones' segmentation and matching, so the LiDAR's static environment is obtained completely by the remained points. By evaluating different types of LiDAR points, the feature-matching error can be alleviated further, making the localization more accurate. Together with qualitative and quantitative results, this work provides a moving objects' detection improvement of 14.13% mAP with the new NeuroIV dataset and an obvious localization accuracy improvement with LiDAR points' evaluation.**

***Index Terms*— LiDAR features' evaluation, localization improvement, movement detection, neuromorphic vision (NeuroIV) sensor, sensors' fusion model.**

I. Introduction

FOR the vehicle's autonomous navigation systems, precise localization is an important component to accomplish the path planning, trajectory tracking control, and navigation to its goal position [1], [2]. Now, the LiDAR and camera have been used widely for environment perception [3]. These AI sensors have been exploited to extract environmental landmarks, and the landmarks' matching is applied to achieve the car's self-localization [4], [5]. However, there are inevitably many moving objects in the actual environment, such as pedestrians, cars, and bicycles [6], [7]. The moving objects' appearance makes the wrong feature-matching with the map's stable features, which can lead to a serious drop in the localization accuracy. Therefore, due to the dynamic disturbance or noise in the scene, it still remains a great demand to detect the moving objects robustly and evaluate different types of laser points in the dynamic scene.

For the detection of moving objects, the camera can provide plentiful visual information for the objects' recognition with the deep learning methods [8]. After the objects are recognized, the movement character can be predicted by the methods, such as the frame difference, optical flow, and so on [9]. The detection methods are mostly based on RGB cameras. Since the movement detection requires multiple frames, the algorithm is complex without instant detection and high robustness. In addition, when the object moves quickly, the RGB camera's image may be blurred, and the prediction accuracy will drop [10]. Unlike the frame-based RGB cameras, neuromorphic vision (NeuroIV) does not have the concept of a "frame" [11] and is insensitive to light. When the objects move, the environmental pixels change, and the NeuroIV generates event points. Furthermore, the NeuroIV has the advantages for movement detection with very little demand for events' storage and computational resources [12], [13]. However, the NeuroIV appears within a short history in the last decade, and it is susceptible to the event noises from the environment, especially when the camera is in the moving status [14], [15]. Therefore, the traditional image processing methods on NeuroIV are vulnerable easily to the NeuroIV's irregular and noise event points from the environment [16]. The real-time detection of moving objects based on NeuroIV needs to be strengthened.

With the purpose of obtaining static points and accurate laser localization results in the dynamic scene, the moving objects should be labeled as noise points from the LiDAR's whole laser points. In addition to that, due to the LiDAR points' sparseness and multiscale characteristics [17], the LiDAR points' extracted features have a certain distribution error. If there are recorded features at a certain moment, each feature owns two constraints (direction and distance),

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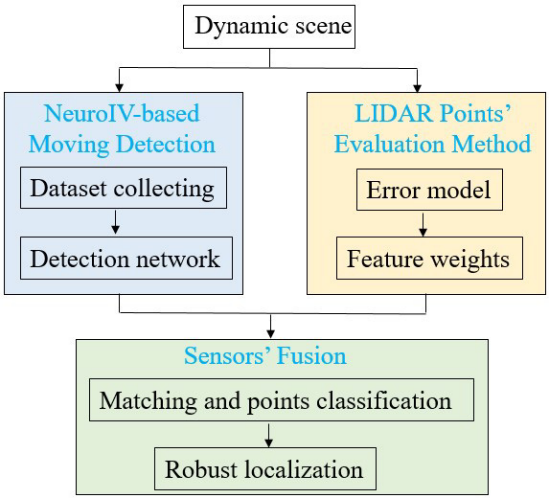


Fig. 1. Framework of the proposed methods.

and the whole constraints' number in the environment will reach $2n$ [18]–[21]. Thus, there will be reliable and unreliable reference problems, which will lead to the LiDAR points' matching and localization uncertainty. Furthermore, the reliability of the same landmark feature will change with the variable observation, and the different feature dependence will lead to different localization accuracies [22], [2]. Therefore, in order to improve the laser localization accuracy, it is still necessary to establish an effective calculation tool or method to evaluate different types of points, including the dynamic points, static points, reliable points, and unreliable points.

To address the above laser localization challenges or requirements from LiDAR and vision-based methods in the dynamic scene, we aim at filling in the gap as few previous works conclude an efficient and robust moving objects' detection solution in a deep learning way with the NeuroIV vision sensor, and the LiDAR's different types of points should be evaluated for robust localization. The framework is illustrated in Fig. 1. The main contribution lies in the following.

1) For the moving objects' detection, a new database is presented, which is the largest dataset of annotated NeuroIV objects with a wide variety of NeuroIV categories. Our proposed YOLO-small network outperforms excellent performance with the new dataset.

2) For the points' evaluation method, the LiDAR points' stereo error model and information entropy tool are established to evaluate different LiDAR points' feature-matching weights, including the dynamic points, static points, reliable points, and unreliable points.

3) For the LiDAR point's static features acquisition, the fusion of NeuroIV and LiDAR is obtained by the method of object zones' segmentation and similar distribution matching. The localization accuracy is improved greatly.

II. Related Work

A. NeuroIV-Based Moving Objects' Detection

Despite that the advent of RGB sensors comes with richer information in pixels' presentation, the NeuroIV camera shows great superiority, especially for the movement capture. Some research works have been attempted to enhance the performance based on that young sensor. The methods can be mainly divided into two categories. The first one is the conventional clustering and feature extraction with later machine learning techniques to achieve classification and regression. Inspired by the mean-shift tracking [23], Comanicu *et al.* [23] created the event clusters by assigning NeuroIV events to the nearest object centroid. Mueggler *et al.* [24] estimated object's posture by a particle filter and gave the moving objects' cluster estimation by NeuroIV camera [24]. Lagorce *et al.* [11] extracted the features by the Hough transform and then classified the event points through the principal component analysis (PCA) method. However, most of the above clustering methods are based on the traditional image processing method, which is susceptible easily to the moving objects' different speeds and event points' thickness. Especially when the object moves at a slow speed or owns a similar pigment, the low event points' thickness may lead to mutilated and nondisplay areas of the moving object, which can only give a partial and incomplete appearance to the whole body.

The other one is the end-to-end deep learning method, which can learn the optimal features from a large amount of well-annotated data. The image's deep learning methods can be roughly divided into two categories: region-based method and end-to-end method. Generally, the region-proposal method' principle is to generate the proposal boxes, such as the sliding window, anchor box, or multiscale combinatorial grouping (MCG) [25]. As to the region-proposal method, the convolutional neural network (CNN) was designed to achieve the objects' recognition in the early stage [26], [27]. To optimize the detection anchors, a perspective-aware methodology was proposed, which divided the image into key regions via clustering [28]. R-CNN is on behalf of the CNN's another milestone, and it uses a selective search method to generate the proposal boxes [29]. Then, a breakthrough method was put forward in Fast R-CNN, which combines the classification and bounding box together [30], [31]. YOLO is a representative of the end-to-end approach, which attempts to discard the objects' proposal regions or boxes [32]. The above detection methods are mostly based on RGB cameras. However, compared to RGB computer vision, NeuroIV, as a young and smaller research field, is rarely associated with the intelligent car. Some detection focuses only on the special scenario, such as the highway and safe driving [33]. Unlike the abundant VOC or COCO RGB datasets used in land or aerial detection [31], there is a scarcity of the NeuroIV vision's dataset. Besides, the basic neural network architectures are largely similar to those networks used in RGB [35]. These RGB-based networks are heavy-weight and computationally time-consuming to process more pixels, which do not take advantage of the only event characteristics in NeuroIV image's pixels.

B. LiDAR Points' Evaluation

For the evaluation method of laser points' reliability, some progress has also been achieved in recent years. Abeg *et al.* [36] analyzed the factors that affected the accuracy of laser localization, including air humidity, particles, light wavelength, and so on. Koivumaki *et al.* [37] pointed out that

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the position of the laser point could be any position in the projected laser spot, which shows the uncertainty of the point position. Schaer *et al.* [38] studied the projection process of the laser point and found that the laser point approximately obeyed the Gaussian distribution. Abeg *et al.* [36] analyzed the influence of the laser emitting hole and the scanning distance on the spot's diameter. Hao *et al.* [40] pointed out that the incidence angle affected the laser spot's long axis length but did not affect the laser spot's short axis length. Chen *et al.* [41] established a laser spot's area model through the incident angle and divergence angle. However, the above research lacks an accurate evaluation of LiDAR points' extracted features. There is no feature's effective error analysis or evaluation model, especially in relation to the laser points' matching process, which has greatly restricted the LiDAR's localization application further.

C. Sensors' Fusion

After the moving objects are detected by the NeuroIV sensor, the detected results need to be correlated with the LiDAR's dynamic zones. For the multisensors' data fusion, the sensors' calibration and time synchronization are the general methods to achieve the fusion. The basic method of synchronization is given in the literature [42]. The method focuses on temporal ordering, where the data from all sensors are ordered chronologically. However, the data frequencies obtained by various sensors are different, and it is difficult to synchronize the time and deduct the moving objects' points [43].

In view of this, some scholars try to optimize the calibration relationship between the two coordinate systems to reduce the influence of matching drift caused by inconsistent sampling frequencies [44]. Bell *et al.* [45] established a sampling frequency model to perform an overall linear interpolation movement of the image within the sampling time interval; thereby, the timestamp can be unified to the same time. Liu *et al.* [46] extracted the inherent rectangular line feature of the LiDAR points in a specific scene, which can be used as a reference for the transformation of different coordinate systems. However, in the dynamic scene, the objects may move quickly; there may be another interpolation drift during the LiDAR and camera's sampling process, which will influence the interpolation compensation.

Then, some scholars have tried to establish a similar relationship between the two information features to promote the matching robustness [47]. Castorena *et al.* [48] converted laser points and images into depth maps, extracted edge and line-surface feature sets, and then established the correspondences relationship. Ding *et al.* [49] achieved the image and points matching by extracting right-angle features. As the abovementioned matching method is based on similar features' extraction, it is susceptible to the blurred image under objects' fast movement. Therefore, it is still necessary to establish an unchangeable and similar matching strategy to suppress the blurring problems in the dynamic scenes.

III. Dataset

A. System Setup

Dataset is the prerequisite for the deep learning method. The first data recording platform is an equipped vehicle in Fig. 2,



Fig. 2. Data recording platform of TjAV and movable bracket for different scenes.

which is capable of fully intelligent driving named Tongji Autonomous Vehicle (TjAV). TjAV is refitted with different kinds of sensors. To build the NeuroIV dataset, the DAVIS346 is used as our main recording sensor, which has a resolution of 346 × 260 pixels with a time resolution of 1 μs. The Velodyne 16-beam LiDAR is placed above the top of TiAV's bracket, with the camera before the fixed bracket. Behind the fixed bracket, a differential global localization system (DGPS) and Oxts RT2000 GNSS provide accurate global localization. Furthermore, in order to enrich the pedestrian datasets, another movable bracket with sensors is placed on the shopping street.

B. NeuroIV-Based Dataset for Moving Objects' Detection

1) Data Encoding and Filter: The NeuroIV camera aims at capturing the moving objects in the environment, and then, the static environment can be obtained. Specifically, each incoming event $[t, x, y, p]$ will appear on the (x, y) pixel at the time stamp t. The event points are projected on the image plane. In Fig. 3(a), due to the different moving speeds, the NeuroIV captures the pedestrian and bicycle with different event points' numbers and thicknesses. Besides, being affected by the environment, there are noise points on the image plan. The image plane with noise points may enlarge the moving object's zone seriously, which can bring challenges to predict the moving objects.

Fig. 3(b) shows the event points' accumulated number or thickness, which is related to the movement speed and color-changing rate within a very short time period. In (1), $n(x,\ y)$ denotes the event thickness at coordinate $(x,\ y)$, and n_T denotes the threshold to distinguish the body and noise events. As the objects' different moving speeds, the threshold (n_T) changes in different dynamic scenes. The Otsu method is designed to calculate the threshold and achieve the filtering operation

$$\text{Otsu}: f(x,y) = \begin{cases} 0, & n(x,y) < n_T \\ 255, & n(x,y) > n_T. \end{cases} \tag{1}$$

It is supposed that there are L levels' events' thicknesses at a certain moment, and all the event points' summaries are n

$$n = n_0 + n_1 + \cdots + n_{L-1} \tag{2}$$

where P_i denotes the probability of the event point number n_i

$$P_i = \frac{n_i}{n}. \tag{3}$$

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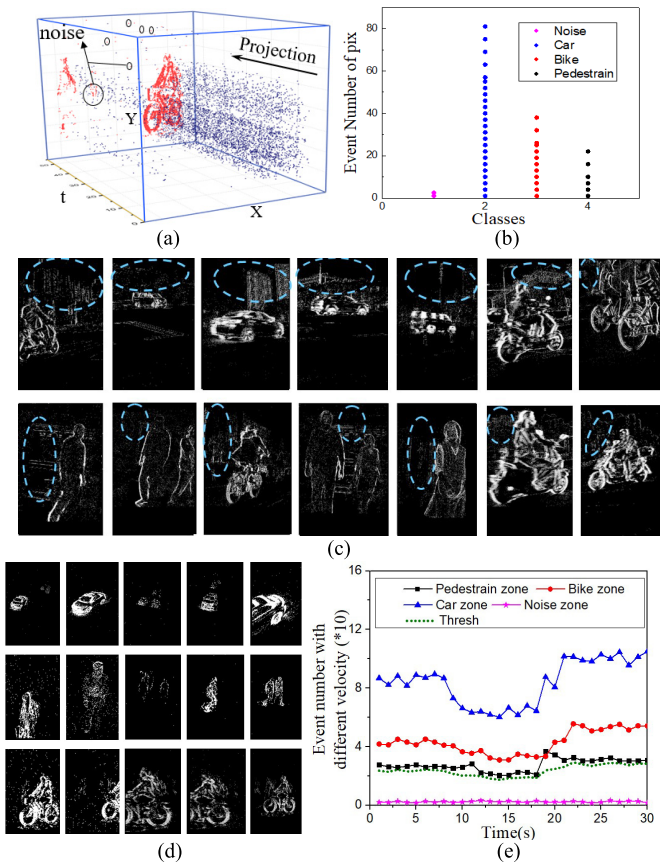


Fig. 3. NeuroIV-based dataset. (a) NeuroIV event points clustering with the fixed time interval. (b) Different type's event points' thicknesses at a time. (c) Moving camera owns more environmental noise circled by ellipses. (d) Static camera owns less environmental noise, including cars, pedestrians, and bicycles. (e) Noises' filtering threshold with the objects' different moving speeds.

Through the probability and the point's number, the average event points' level (u_T) is calculated as follows:

$$\mu_T = \sum_{i=0}^{L-1} n_i.P_i. \tag{4}$$

The events on the NeuroIV image are divided into moving object's events and noise events. In (5), $\omega_0(n_T)$ and $\omega_1(n_T)$ denote the noise and body events' probability, respectively, which are distinguished by the event points' thickness threshold (n_T)

$$\begin{cases} w_0(n_T) = \sum_{i=0}^{T} P_i \\ w_1(n_T) = 1 - w_0(n_T). \end{cases} \tag{5}$$

Equation (6) shows the average number of event thickness in the body and noise zones

$$\begin{cases} \mu_0(n_T) = \dfrac{\sum_{i=0}^{T} n_i.P_i}{w_0(n_T)} \\ \mu_1(n_T) = \dfrac{\mu_T - \sum_{i=0}^{T} n_i.P_i}{w_1(n_T)}. \end{cases} \tag{6}$$

In (7), $\sigma^2(T)$ denotes the events' thickness variance of the body and noise zones

$$\sigma_B^2(n_T) = w_0(n_T)[\mu_0(n_T) - n_T]^2 + w_1(n_T)[\mu_1(n_T) - n_T]^2. \tag{7}$$

$n_T{}^*$ is calculated finally when the thickness variance reaches maximum

$$\sigma_B^2(n_T^*) = \max_{0\le T\le L-1}\{\sigma_B^2(n_T)\}. \tag{8}$$

Each pixel's event points' thickness is compared with the threshold ($n_T{}^*$), and the noise or body pixels are distinguished by (1). Fig. 3(c) shows that there is a large amount of environmental noise in the camera's moving status. Fig. 3(d) shows that there are fewer environmental noises in the camera's static status. As the moving objects can lead to the pixels' harsh changes rate, they own obvious event thickness threshold than the environmental noises. Through the designed Otsu method, the moving objects' event body features are retained, and the noise points can be filtered. The robustness of the OTSU encoding method can be well verified in Fig. 3(e). Although there may be mixed objects with different speeds, such as pedestrians, and vehicles, the OTSU shows excellent performance in the thresholds' calculation and objects' distinguishment under complex environments.

2) Data Classes: According to the filtering algorithm of OTSU encoding, we record five frames per second, which can ensure all moving speed frames without similar recordings. By doing so, we are able to cover the whole datasets' information.

Visualization of samples from the moving objects' dataset is shown in Fig. 3(d). The NeuroIV dataset consists of three subdatasets: the pedestrian dataset, the bicycle dataset, and the car dataset. Each image consists of 346 × 260 pixels and contains objects with different scales, orientations, and shapes. These NeuroIV images are then annotated with arbitrary quadrilaterals. We gather data from five different scenarios in 30 situation cases, including the street, campus, shop entrance, and so on. All moving objects either go straight or turn at different speeds. A summary of the NeuroIV dataset is shown in Table I.

3) Annotation: In total, we annotated 38 522 frames of NeuroIV images. About half of them, 17 150 annotated frames, are mixed with other moving targets. The average annotation box size is 1.4 cm in width and 2.5 cm in length. The data are first sparsely sampled to 5 Hz to keep movement inconsistency of data sources and then converted into .png format files for hand labeling. The final annotations are given in plain text files, and each corresponds to its frame file. Every element in [C, l, ω, xc, yc] row represents an annotated target box with its class C, size in length, width, and center coordinates of xc, yc.

C. LiDAR Points' Datasets in the Dynamic Scene

The robot operating system (ROS) is used to communicate between LiDAR and industrial personal computer (IPC). Therefore, ROS bags are saved to record ROS topics, and

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TABLE I
NEUROIV DATASET STATISTICS

Datasets	The numbers of NEUROIV sample	Scenes	Movement	Speed
pedestrian	15500	street, campus and shop	Line and turning movement	10-50m/min
bicycle	16400	street, campus	Line and turning movement	50-300 m/min
car	17300	street, campus	Line and turning movement	0.5-2.0 km/min

LiDAR points' topics can be replayed in the format of rosbag. Each bag file contains a header topic of timestamps, laser angle, and laser distance informatics from a Velodyne LiDAR as the topic "velodyne_points."

Dynamic scenarios are mainly recorded in real road driving scenes, including campus streets, indoor venues, and city roads. Intersections, road centers, and so on are used as the collection places, which includes pedestrians, moving vehicles, and other objects (Fig. 4) [50]. In real traffic, we record adverse traffic flow, turns, and U-turns. At the same time, there are many static factors in the environment as localization references, including roadside buildings, bushes, columns, and so on.

IV. MATERIALS AND METHODS

A. Moving Objects' Detection Based on the YOLO-Small Network

After the NeuroIV-based dataset is built, a YOLO-small network is designed to detect the moving objects based on the NeuroIV-annotated dataset. Targets in the NeuroIV image can be very sparse on various scales, especially when the objects move at a slow speed. Also, there may be mutilated and nondisplay areas for different types of moving objects. For example in Fig. 3(d), objects appear only partially due to the speed inconsistency or pixel similarity. As the moving objects (car, pedestrian, and bicycle) own fixed and stable profiles, it can also provide a great possibility for the network to recognize them from the complicated environmental interferences. Thus, it is necessary to predict the whole body.

To this end, we propose a YOLO-small network to improve the detection robustness, which utilizes a one-stage network with a stacked feature map structure. The detection network includes two parts: the backbone layer and the head layer. The YOLO-small network is a simplified frame, whose head layer is consulted from the YOLO network. The YOLO-small's detection network is drawn in Fig. 5.

1) Backbone Layer: The encoded images of NeuroIV are extracted to form a feature map pyramid. Through this pyramid, regions of interest (ROIs) are proposed by learning from offsets and errors between proposal boxes and matched ground truths.

To be more specific, in the first feature map pyramid structure, the residual network is used as the backbone feature extractor unite over the NeuroIV dataset. Through every feature extraction layer, the NeuroIV image is downsampled in size and gradually expanded to 256 channels. Four

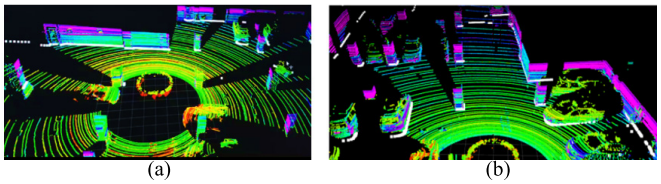


Fig. 4. LiDAR points' dataset in the dynamic scene with static and moving objects. (a) Dynamic scene in the outdoor. (b) Dynamic scene in the indoor.

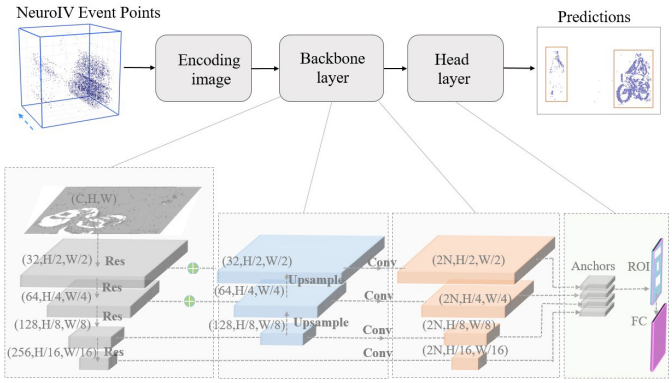


Fig. 5. YOLO-small detection network with the NeuroIV dataset.

specific-sized feature maps among them are concatenated together to form a four-layer top-down feature map pyramid with progressively decreasing resolutions.

A single convolution operation is conducted on the (H/8, H/8, 128) feature layer. After that, the deconvolution and upsampling conductions are combined with other feature layers. Note that, at each upsampled layer, the feature map is added by the corresponding feature map from the previous pyramid. In this way, the feature accuracy can be improved by the increased depths. In addition, the upsampling jump connection can alleviate the gradient disappearance problem, which is caused by the deepened layers.

The third pyramid is derived from the previous two feature pyramids. By the Conv operation, the previous pyramid layer provides four layers of feature maps. Later, in the third pyramid, anchor boxes are generated simultaneously and independently within each layer, capturing proposal objects of different sizes and scales. Anchors that successfully match the ground truth will guide the network to generate more suitable proposed boxes by learning their offsets and errors in binary classification, box size, and box orientation. Proposals are then concatenated together.

2) Head Layer: The fourth pyramid structure is utilized in the RoI detection head. Each proposal box cuts out its RoI feature map from the third pyramid. Final detection of every target generates upon the aggregated RoI zones. Fully connected layers flatten the last feature map and output the vector encoding desired results. The output will be overlaid back onto the original NeuroIV-image, and NMS calculation can be updated and back propagated before the final losses of classification, box size, and orientation.

The output layer's size is (H/2, W/2; H/4, W/4; H/8, W/8; and H/16, W/16). The detection is based on the NeuroIV's event points and three labels (pedestrian, bike, and car).

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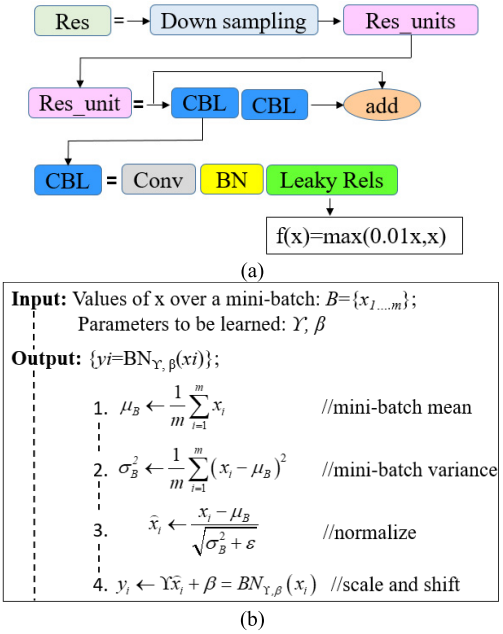


Fig. 6. ResN structure in the YOLO-small network. (a) ResN structure's information diagram. (b) BN's batch normalization.

YOLO-small has three preverification boxes, and each prediction box includes five parts: x offset, y offset, length, width, and trust index. Thus, the last output layer's dimension is $3^*(5+3) = 3^*8 = 24$.

3) Feature Map Unit: The Res structure is composed of Conv, BN, and leaky sublayers. The information diagram among them is drawn in Fig. 6. The BN layer performs the following transformations.

1) Normalize the feature data of each batch sample, respectively, so that each feature data can be transformed into a similar distribution with the scope of 0 mean value and 1 variance. The purpose of the conversion is to train data conveniently within a uniform scope.
2) Make $yi = Yxi + \beta$, and conduct another linear transformation on the output data of the previous step. The linear transformation operation can restore the feature to its original scale.

4) Implementation:

a) Data augmentation: The NeuroIV images are filtered and encoded at regular time intervals. Then, they are converted to an RGB pattern for better compatibility with YOLO pretrained weights. Every ground-truth box is randomly annotated.

b) Network settings: The input image of this network has a resolution of 346 × 200. The generation and matching strategies of anchor boxes are given as follows: each anchor box is generated by width, length, and center point. The network's anchors own the same aspect ratio as the YOLO-V5 network, but their sizes are scaled to cater to NeuroIV's image size. The anchors' sizes are [5, 7; 9, 16; 17, 12], [16, 32; 33, 24; 31, 64], and [62, 48; 84, 106; 137, 37]. A 1:1 positive-to-negative ratio is used as the resampling ratio. At inference time, the rotational nonmaximum suppression (NMS) with an overlap threshold of 0.3 intersection over union (IOU) is used to select the optimal anchor. This achieves a similar detection speed but owns better performance than axisymmetric NMS.

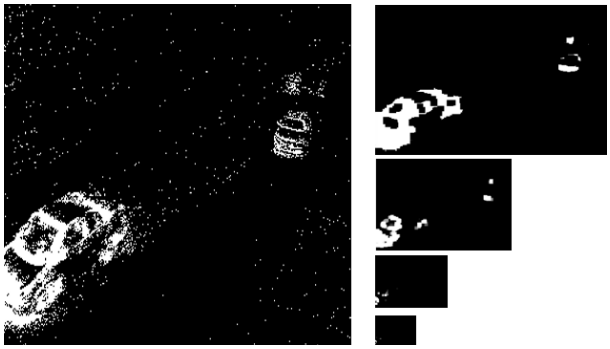


Fig. 7. Visualized feature maps with the Yolo-small network. (a) Original objects that are captured without filtering preparation. (b) Feature maps from the backbone structure in four scale levels.

The stochastic gradient descent (SGD) with momentum is used for network training. Two GTX 3080 GPUs are used to run the optimizer with a total of 12 frames per minibatch. Models are trained for 150 epochs at maximum. The initial learning rate is 0.0005, with an exponential decay factor of 0.8 and a decay every 15 epochs. A weight decay regularization of 0.0005 and a momentum beta value of 0.9 are used. It takes ∼24 h to train the moving object detection network on two GTX 3080 GPUs.

c) Loss function: The loss function of the prediction box includes six parts: the prediction center's x coordinate, y coordinate, width, height, IOU error, and the prediction classes. The prediction's loss function is defined as follows:

$$\begin{aligned}\text{Loss} &= \lambda \sum_{i=0}^{s^2}\sum_{j=0}^{2} \|_{ij}^{\text{obj}} \left[\left(x_i - \hat{x}_i\right)^2 + \left(y_i - \hat{y}_i\right)^2\right] \\ &+ \lambda \sum_{i=0}^{s^2}\sum_{j=0}^{2} \|_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i}\right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i}\right)^2\right] \\ &+ \sum_{i=0}^{s^2}\sum_{j=0}^{2} \|_{ij}^{\text{obj}} \left(C_i - \hat{C}_i\right)^2 + \sum_{i=0}^{s^2}\sum_{j=0}^{2} \|_{ij}^{no\text{obj}} \left(C_i - \hat{C}_i\right)^2 \\ &+ \sum_{i=0}^{s^2} \sum_{c\in\text{class}} \|_{ij}^{\text{obj}} \left(p_i - \hat{p}_i\right)^2. \end{aligned} \tag{9}$$

To obtain features during extraction, the extracted feature map is shown in Fig. 7, which shows that the network has learned to infer dense points' clusters from raw corners and edges. It is reasonable and understandable for these extracted features because they conform to the loss constraints.

B. LiDAR Point's Evaluation Method

In order to evaluate different types of LiDAR points, it is necessary to establish an evaluation method. The extracted line and surface features from the LiDAR points own more robustness, which can reduce the complexity of the matching process [54]. However, due to the sparseness of LiDAR's

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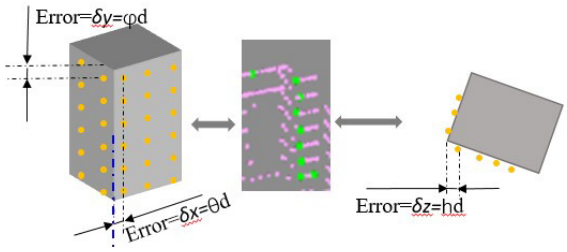


Fig. 8. LiDAR points' error distribution on line and surface features.

points, there are distribution deviations between the extracted feature and the actual feature. The deviations are distributed along with the width, height, and depth directions (in Fig. 8). It is assumed that there is a distribution error vector for the extracted feature. The error vector is $pe = [\delta x, \delta y, \delta z]$. δx is related to horizontal angle resolution θ and scanning distance (d). δy is related to vertical angle resolution φ and scanning distance (d). δz is related to air medium h and scanning distance (d). $\delta x = \theta d = \lambda_1$, $\delta y = \varphi d = \lambda_2$, and $\delta z = hd = \lambda_3$.

The information entropy is defined as the occurrence probability of discrete random events, and it is one of the most important theories in modern measurement [51]. The calculation of information entropy is defined in (10) [52], [53]. $P(X)$ denotes the probability of X event. Therefore, information entropy can be introduced into the evaluation of the feature's distribution uncertainty, which is also a denotation of the distribution error's size or volume

$$\begin{cases} \text{Information entropy}: P = -\sum_{i=1}^{n} p(x_i)\log p(x_i) \\ \text{Error entropy}: S = e^{P}. \end{cases} \tag{10}$$

The points feature's actual distribution rules will influence the calculation results of information entropy, so it is necessary to build the mathematical expression of the error model accurately. If the error distribution obeys a uniform distribution, the error volume appears a cube shape. If the error distribution obeys the Gaussian distribution, the error volume appears in an ellipsoidal shape. The distribution errors and mathematical expressions are shown in Table II. The three axes of the cube or ellipsoid are composed of δx, δy, and δz, respectively.

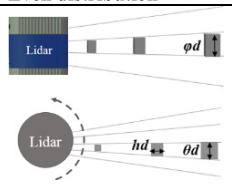
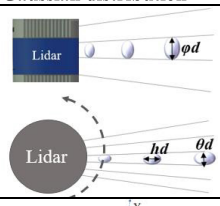
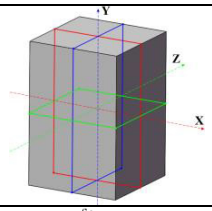
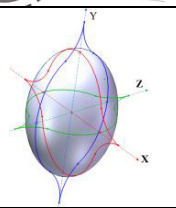
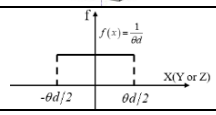
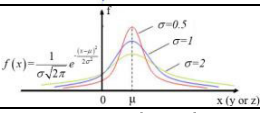
In the even distribution model, there is only one parameter, and $(x) \sim U\ (-0.5\lambda,\ 0.5\lambda)$. In fact, all the stereo points fall into the cube's ranges: $[-0.5\lambda 1,\ 0.5\lambda 1]$, $[-0.5\lambda 2,\ 0.5\lambda 2]$, and $[-0.5\lambda 3,\ 0.5\lambda 3]$. Since the total probability integration of the error cube is 1, (11) can be obtained. Then, (12) can be simplified further. According to the definition in (10), the error entropy of the even distribution model can be calculated in (13). Through the error entropy, it can be easily found that the detection accuracy changes linearly with the scanning distance

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z)dx_dy_dz = 1 \tag{11}$$

$$\begin{aligned} P_{\text{even}} &= -\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z)\ln[f(x,y,z)]dx_dy_dz \\ &= -\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z)\ln\left(\frac{1}{\lambda_1\lambda_2\lambda_3 d^3}\right)dx_dy_dz \\ &= \ln\left[\lambda_1\lambda_2\lambda_3 d^3\right].\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z)dx_dy_dz \\ &= \ln\left[\lambda_1\lambda_2\lambda_3 d^3\right] \end{aligned} \tag{12}$$

$$S_{\text{even}} = e^{P} = \lambda_1\lambda_2\lambda_3 d^3 = 1.84 * 10^{-7} * d^3. \tag{13}$$

TABLE II
ERROR MODEL OF POINTS' DISTRIBUTION

	Even distribution	Gaussian distribution
Stereo error model		
Points' error model of VLP LIDAR.		
Distribution curves		
Distribution function	$f(x)=\frac{1}{\delta_x}=\frac{1}{\theta d}$ ⇓ $f(x,y,z)=\frac{1}{\delta_x.\delta_y.\delta_z}$ $=\frac{1}{\theta\varphi h.d^3}$	$f(x)=\frac{1}{\lambda(2\pi)^{1/2}}\exp\left[-\frac{1}{2}\frac{x^2}{\lambda^2}\right]$ ⇓ $f(x,y,z)=$ $\frac{1}{(2\pi)^{3/2}\lambda_1\lambda_2\lambda_3}\exp\left[-\frac{1}{2}\left(\frac{x^2}{\lambda_1^2}+\frac{y^2}{\lambda_2^2}+\frac{z^2}{\lambda_3^2}\right)\right]$

In the Gaussian distribution model, there is only one parameter, and $(x) \sim N\ (0,\ \lambda^2)$. In fact, 97% of the LiDAR points fall into the ellipsoid's ranges: $[-\lambda 1+\mu,\ \lambda 1+\mu]$, $[-\lambda 2+\mu,\ \lambda 2+\mu]$, and $[-\lambda 3+\mu,\ \lambda 3+\mu]$.

Make $x' = x/(\sqrt{2}\lambda_1)$, $y' = y/(\sqrt{2}\lambda_2)$, $z' = z/(\sqrt{2}\lambda_3)$; (14) can be simplified further as (15)

$$\begin{aligned} P_{\text{gaussian}} &= -\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z) \\ &\quad\times\left[\ln\left(\frac{1}{(2\pi)^{3/2}\lambda_1\lambda_2\lambda_3}\right)\right. \\ &\qquad\left.+\ln\left(\exp\left[-\frac{1}{2}\left(\frac{x^2}{\lambda_1^2}+\frac{y^2}{\lambda_2^2}+\frac{z^2}{\lambda_3^2}\right)\right]\right)\right]dx_dy_dz \\ &= \ln\left(\frac{1}{(2\pi)^{3/2}\lambda_1\lambda_2\lambda_3}\right)\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z)dx_dy_dz \\ &\quad+\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y,z) \\ &\quad\cdot\left(\frac{x^2}{\left(\sqrt{2\lambda_1}\right)^2}+\frac{y^2}{\left(\sqrt{2\lambda_2}\right)^2}+\frac{z^2}{\left(\sqrt{2\lambda_3}\right)^2}\right)dx_dy_dz \end{aligned} \tag{14}$$

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$$P_{\text{gaussian}} = \ln\left[(2\pi)^{\frac{3}{2}}\lambda_1\lambda_2\lambda_3\right] + \frac{1}{\pi^{\frac{3}{2}}}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\left(x'^2+y'^2+z'^2\right) \times \exp\left[-\left(x'^2+y'^2+z'^2\right)\right]dx'dy'dz'. \quad (15)$$

By the ball area integration and step integration methods, (15) can be transformed into (16) [41]. According to the definition between the information entropy and error entropy in (10), the Gaussian distribution's error entropy can also be calculated in (17)

$$\begin{aligned} P_{\text{gaussian}} &= \ln\left[(2\pi)^{\frac{3}{2}}\lambda_1\lambda_2\lambda_3\right] + \frac{1}{\pi^{\frac{3}{2}}}\iiint_{\Omega} r^2\exp\left(-r^2\right)r^2(\sin\theta)drd\theta d\phi \\ &= \ln\left[(2\pi)^{\frac{3}{2}}\lambda_1\lambda_2\lambda_3\right] + \frac{4}{\sqrt{\pi}}\int_0^{+\infty} r^4.\exp\left(-r^2\right)dr \\ &= \ln\left[(2\pi)^{\frac{3}{2}}\lambda_1\lambda_2\lambda_3\right] + \frac{3}{2} \quad (16) \\ S_{\text{gaussian}} &= e^P = (2\pi)^{\frac{3}{2}}\lambda_1\lambda_2\lambda_3 e^{\frac{3}{2}} \\ &= (2\pi e)^{\frac{3}{2}}.(\theta\phi h).d^3 \\ &= 1.296^{*}10^{-5^{*}}d^3. \quad (17) \end{aligned}$$

C. Sensors' Fusion by the Graph Nodes' Batch-Based Matching

The LiDAR's points are composed of ground points and spatial points. The points' normal vector calculation method is utilized to strip the LiDAR's ground points. In fact, points with a vertical normal vector are assumed to be the ground point. Set the threshold h; if the point's normal vector is less than h, it is judged as the ground point; otherwise, it is the spatial point. The ground filtering method by normal vector threshold can separate the ground points from the spatial points quickly and effectively.

After that, the LiDAR points remains only the objects' spatial points, which have relatively good independence. The Euclidean distance is a commonly used metric among spatial points, and the Euclidean distance is used to quickly segment different objects. Thus, different objects' spatial segmentation can be achieved in the LiDAR points' world.

Due to the sensors' different sampling frequencies and the objects' rapid movement, there will be a matching drift between the NeuroIV detection and LiDAR points. To address the drift problem, we adopt a batch matching strategy by the graph nodes.

On the basis of points' segmentation and NeuroIV objects' prediction, the distribution zone models of objects in the two modes are extracted, respectively. The number of distribution zones of NeuroIV objects is less than the number of object distribution zones in points' segmentation. As shown in Fig. 9, similar correspondences between model edges and angles in the two distributions are established, respectively, which can be used to achieve objects' zone matching and dynamic points' detection.

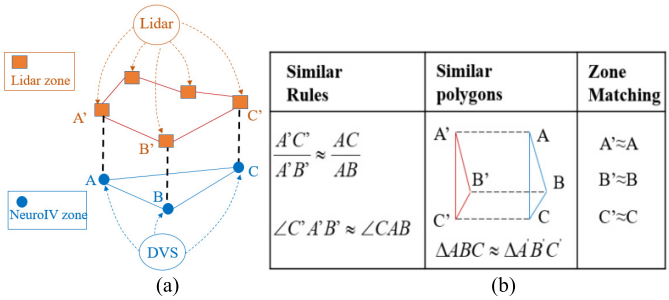


Fig. 9. LiDAR and NeuroIV's fusion process. (a) Objects' distribution zones in the LiDAR and camera modes. (b) Different zones' matching relationships with similar rules.

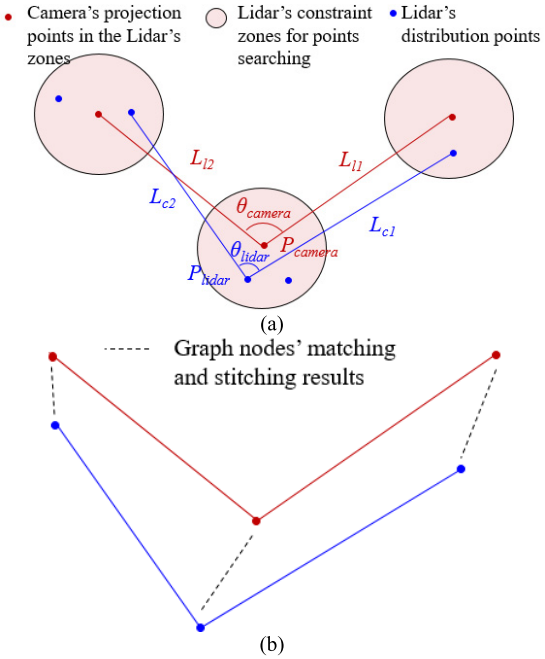


Fig. 10. Graph nodes' projection, searching, and stitching processes. (a) Searching process and principles. (b) Graph nodes' stitching results.

By calibrating the coordinate systems between the NeuroIV and LiDAR sensors, the distribution zones projected from the graph nodes in NeuroIV are established. Constrained by the zones' searching radius, the matching relationship between the NeuroIV detection and the LiDAR points' segmentation on the spatial distribution can be achieved.

As shown in Fig. 10, the moving objects detected by the NeuroIV are projected into the LiDAR's spatial zones. A descriptor (S_q) is utilized to denote the graph nodes within the zones. $S_q = (P, \theta, E)$. Among them, P denotes the distribution position of the node, θ denotes the angle of the node's connection lines, and E denotes the line length ratio of the node. Equation (18) is utilized to calculate the similarity between the detected node (Q_{lidar}) in the LiDAR and the projected node (Q_{camera}) from the NeuroIV. Equation (19) is utilized to define the node's line length ratio. Among them, S_P denotes the position correlation coefficient, S_θ denotes the angular similarity coefficient, and S_E denotes the similarity coefficient of the node's line

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length ratio

$$S_q\left(Q^t_{\text{lidar}}, Q^t_{\text{camera}}\right) = S^*_P\left(P^t_{\text{lidar}} - P^t_{\text{camera}}\right) + S^*_\theta\left(\theta^t_{\text{lidar}} - \theta^t_{\text{camera}}\right) + S^*_E\left(E^t_{\text{lidar}} - E^t_{\text{camera}}\right) \quad (18)$$

$$E_{\text{lidar}} = \frac{L_{l1}}{L_{l2}}, \quad E_{\text{camera}} = \frac{L_{c1}}{L_{c2}}. \quad (19)$$

Equation (20) is utilized to calculate the association value between the nodes, and (21) is utilized to obtain the best similarity relationship. Among them, N denotes the number of nodes in one constraint zone, and M denotes the number of constraint zones

$$Q^t = \exp\left(s_q\left(Q^t_{\text{lidar}}, Q^t_{\text{camera}}\right)\right) \quad (20)$$

$$\Gamma^t = \arg\min \sum_{q=1}^{M}\sum_{i=1}^{N} Q^t. \quad (21)$$

By stitching the graph nodes detected by the NeuroIV and LiDAR sensors, the calibration relationship between the two sensors can be improved greatly, and the moving objects in the LiDAR points can be accurately deducted.

D. Different Points' Evaluation in the Laser Localization Process

When the autonomous vehicle is traveling, there is a map feature set G and current traveling feature set P. Through the rotation and shift matrix (R, H), the feature set P is recalculated as the feature set P_T. According to (22), the least-squares distance error (E_{matching}) from P_T to G is calculated. If the E_{matching} value is less than the predefined threshold, the current feature and map feature match successfully, and the current position is calculated finally by the posture matrix (R, H)

$$E_{\text{matching}} = \text{argmin}_{R,t} \sum_{i=1}^{n} w_i^* \|G_i - (R.P_i + H)\|^2. \quad (22)$$

According to the NeuroIV image's detection and laser features' evaluation, the information entropy of moving objects and static objects are calculated separately. Then, the features' matching weights (w_i) in (22) are optimized. The LiDAR and NeuroIV' fusion localization algorithms are shown in Table III and Fig. 11.

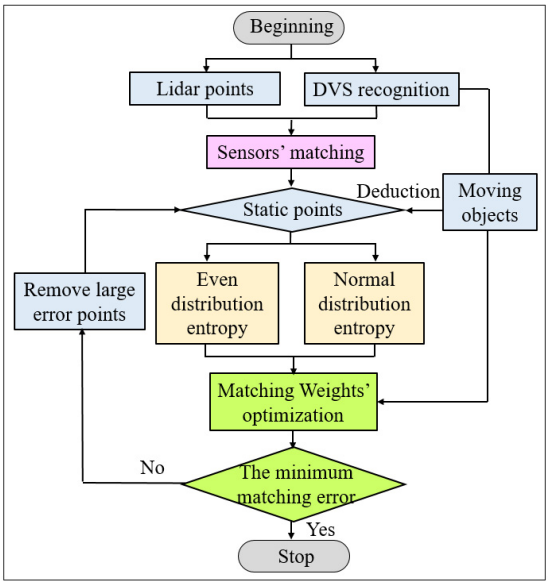


Fig. 11. Whole localization algorithm by the NeuroIV' moving objects' detection and LiDAR points' evaluation.

TABLE III
POINTS' ERROR ENTROPY AND MATCHING WEIGHTS OF DIFFERENT OBJECTS

	Static points		**Dynamic points**
	Even distribution	Gaussian distribution	Noise
Information entropy	$P_{even} = ln[\lambda_1\lambda_2\lambda_3 d^3]$	$P_{gaussian} = ln\left[(2\pi)^{\frac{3}{2}}\lambda_1\lambda_2\lambda_3\right] + \frac{3}{2}$	$+\infty$
Weights' distribution	$PP_i^{even} = 1 - \frac{P_i^{even}}{\sum_{i=1}^{n} P_i^{even}}$ ⇓ $w_{obj,i}^{even} = \frac{PP_i^{even}}{\sum_{i=1}^{n} PP_i^{even}}$	$PP_i^{gaussian} = 1 - \frac{P_i^{gaussian}}{\sum_{i=1}^{n} P_i^{gaussian}}$ ⇓ $w_{obj,i}^{gaussian} = \frac{PP_i^{gaussian}}{\sum_{i=1}^{n} PP_i^{gaussian}}$	0
The final weights	$w_i^{static} = 0.5*w_{obj,i}^{even} + 0.5*w_{obj,i}^{gaussian}$		0

V. EXPERIMENTAL VERIFICATION

A. Moving Detection's Qualitative and Quantitative Comparison

We carried out the experiments over the NeuroIV dataset with the following evaluation metrics: the predictions are accessed by moving objects' centers and classes. Different criteria for distance to object center point and object classification error were tested separately. A detection is considered accurate only if its predicted center offset is below some threshold.

Detection comparisons are shown in Table IV. The Yolo-small achieves around an average precision (AP) of 71.2% with a positive criterion of 0.9 IOU. Comparisons with other detection types prove the effectiveness of the proposed dataset and network. The amplitude of various speeds is largely steady. As for the calculation speed, Yolo-small can achieve 25 ms per frame, above 41-Hz fresh rate, which is an import guarantee for real-time detection.

From the experiment results, intuitively, as the objects' moving speed goes higher, the AP appears to be a significant promotion. This is most likely because faster speed will generate more event points, and the objects' accumulated profiles will be more stable and obvious. Different detecting distances results show that NeuroIV's detection performance

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TABLE IV
COMPARISON WITH OTHER MODELS

Model	IMAGE	Events	Backbone	Head	mAp	FPS
RCNN	RGB, VOC,COCO	Static, dynamic	VGG16	Proposal Two-stage	43.7	3
FastRCNN	RGB VOC, COCO	Static, dynamic	VGG16	Proposal Two-stage	60.0	15
RetinaNet	RGB VOC, COCO	Static, dynamic	ResNet-FPN	ProposalFree One-stage	58.3	35
SSD300	RGB VOC, COCO	Static, dynamic	VGG16	ProposalFree One-stage	64.3	38
YOLO-V3	RGB VOC,COCO	Static, dynamic	Darknet 53	ProposalFree One-stage	56.4	34
YOLO-small	NeuroIV	dynamic	ResNet-FPN	ProposalFree One-stage	71.2	41

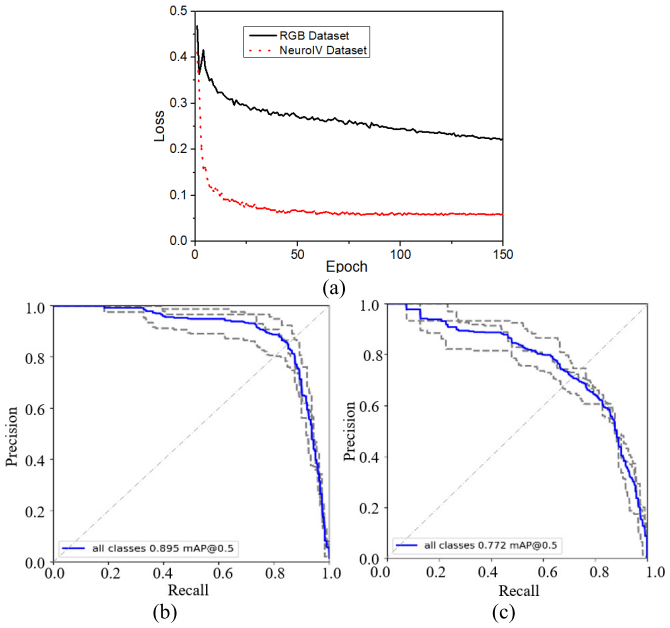


Fig. 12. RGB and NeuroIV's detection comparison. (a) Training loss's comparison between the two different types of datasets. (b) P–R curve by the NeuroIV-based detection, including the car, bike, and pedestrian. (c) P–R curve by the RGB-based detection, including the car, bike, and pedestrian.

gradually increases as the moving objects are closer to the sensor's observation center. The reason could be that far-moving objects appear low area occupation and resolution. We tested distance ranges of (15.56, 14, 9, 6) m in front of the ego-sensor. The results further confirm that it is more suitable for closer detection tasks, such as fast-moving, mixed, and changeable scenes.

In order to further compare the detection effects of different types of cameras, we accomplished the object detection training process under RGB and NeuroIV datasets, respectively. In addition, a little test dataset with a 1:1 inverse ratio is prepared to test precision and recall. The relevant PR curves and training weight files are shown in Fig. 12 and Table V, respectively.

TABLE V
NETWORK'S COMPARISON

Model	mAP(%)			FPS	Training weights' file
	Small	Middle	Large		
YOLO-small	41.3	63	76.8	40	12.1M
YOLO-V3	27.7	49.1	62.9	25	20.8M

Compared to RGB and learning-based methods, NeuroIV-based detection significantly outperforms state-of-the-art effects. The YOLO-small method also achieves competitive results considering it is an ultralight weighted model at just a 12.1-MB training weight file. It can inference at a much higher speed of ∼25 ms per frame, while it still holds state-of-the-art movement capture. The results in Fig. 12 and Table V show that NeuroIV's YOLO-small prediction network has higher detection accuracy and speed.

B. LiDAR and NeuroIV Sensors' Fusion Comparison

Based on the network and training weight files of the NeuroIV dataset, the detection of moving objects in the indoor and outdoor scenes is carried out, respectively, and the spatial distribution models of moving objects are established, respectively. At the same time, the LiDAR points are recorded with the moving objects' appearance, including the pedestrian, moving cars, and so on. The ground LiDAR points are removed by the normal vector calculation. Then, different objects' points' distribution models by the connected domain segmentation are achieved [see Fig. 13(a) and (c)].

After extracting different objects' distributions in NeuroIV and LiDAR points' zones, all distribution zones are connected along the outer contours, which forms two polygons. According to the similar edge and angle laws in different polygons, the similar matching relationship between the image and LiDAR points in the spatial distribution is excavated [see Fig. 13(b) and (d)].

The essential cause for object matching drift is that the sampling frequencies between LiDAR and NeuroIV are inconsistent. The sampling frequency of the LiDAR is within the scope of 5 ∼ 20 Hz, so the maximum time period to record a frame of LiDAR points can reach 0.2 s. In contrast, the processing speed of the NeuroIV camera can reach the level of us. Therefore, when the object is moving rapidly, the same object will appear as a matching drift within the sampling period of the two sensors. To demonstrate the matching drift based on sensors' calibration, the extreme experimental conditions are set and conducted: the slowest rotation frequency of the LiDAR is set to be the 5 Hz, and there are fastest moving vehicles and pedestrians in the environment.

The matching results are shown in Fig. 13(e) and (f). When the car moves fast, the same car cannot be matched and corresponded by the sensors' calibration method. However, the matching accuracy of the pedestrian is higher than that of the car. This proves that the matching drift is related to the sensor's sampling frequency and the object's moving speed. Therefore, when the moving objects' speed cannot be changed

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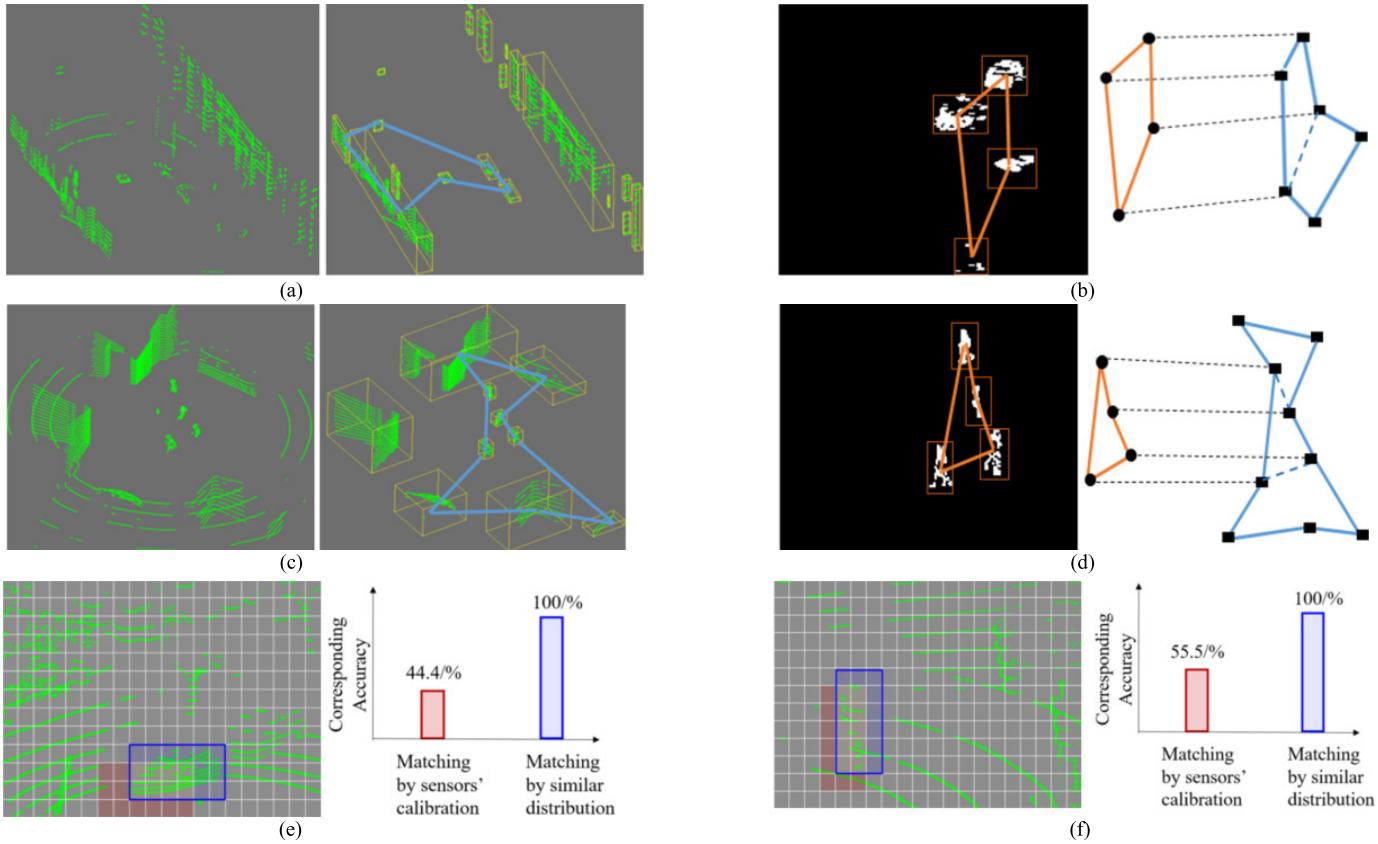


Fig. 13. LiDAR and NeuroIV's matching processes and comparisons in the indoor and outdoor scenes. (a) and (c) LiDAR' ground points' filtering and objects' segmentation. (b) and (d) NeuroIV's objects' distribution and similar matching with the LiDAR's distribution. (e) Extreme experiment: corresponding accuracy for the moving car in the LiDAR points. (f) Extreme experiment: corresponding accuracy for the pedestrian in the LiDAR points.

in the sensors' calibration method, a higher rotation frequency of the LiDAR should be set to reduce the matching error. However, the rotation frequency of LiDAR still has an upper limit.

In contrast, although there may be inconsistent sampling frequency for different sensors, the matching method by similar distribution can overcome the challenge of matching drift. Through the construction of the spatial distribution model and similarity matching, the dynamic points' zone in the LiDAR points can be corresponded and removed completely. This reduces the reliance on the sensor calibration accuracy and also reduces object matching drift when facing the fast movement.

C. Laser Localization Comparison in the Dynamic Scene

To validate the NeuroIV-based moving objects' detection and LiDAR points' evaluation method, it is necessary to test in the dynamic scenes with different elements. We select indoor and outdoor scenes. In the indoor scene, the pedestrians are the moving objects and noise elements for localization. In comparison, there are more noise elements for the outdoor scene, including the bicycles, pedestrians, and moving vehicles with different moving speeds. The NeuroIV camera is utilized separately to achieve moving objects' detection. The moving objects are detected by the YOLO-small network. The environment's whole laser points are recorded by the Velodyne 16-line LiDAR.

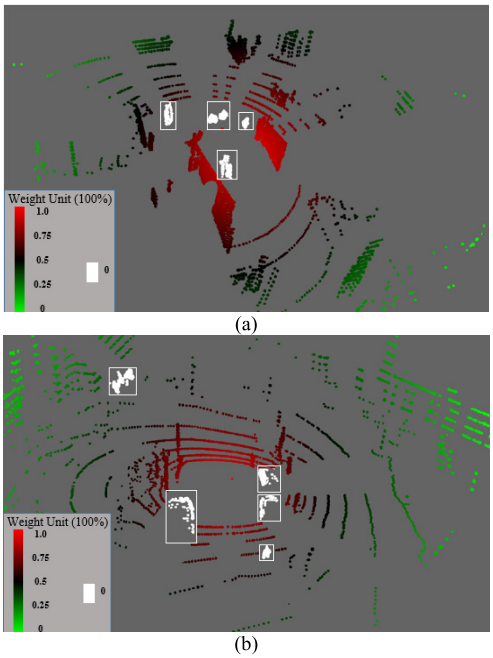


Fig. 14. LiDAR points' feature evaluation with the moving objects' deduction. (a) Indoor scene. (b) Outdoor scene.

After recognizing the moving objects, the matching model is established by the NeuroIV image's prediction boxes and LiDAR points' segmentation. The LiDAR points on the

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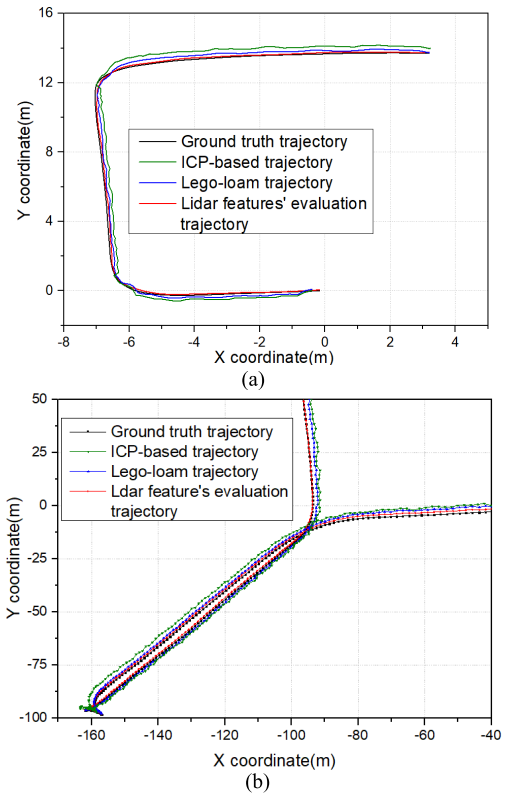


Fig. 15. Multisensors' fusion and localization results. (a) Indoor scene. (b) Outdoor scene.

moving objects are marked as noise points, and their matching weights are set as zero. Then, the NeuroIV's prediction boxes are used to correspond and deduct the environment's dynamic points. According to the information entropy's calculation method, the weights of remaining static buildings are recalculated. The matching weights are recalculated on the remained static features. The recognition and evaluation results by NeuroIV and LiDAR are shown in Fig. 14.

After the matching weights are evaluated, the localization results are optimized. The ICP-based method is the traditional points matching method, which does not conduct the points features' extraction in the localization process. Lego-loam can extract the line and surface features from the environment, and it is a lightweight and ground-optimized LiDAR localization method on variable terrain [54]. Thus, they are utilized to achieve the comparison for the ground-truth localization. The optimized localization results are compared in Fig. 15.

The comparison shows that the combination of NeuroIV-based moving objects' detection and LiDAR points' evaluation has higher localization accuracy. In comparison, the NeuroIV only detects the moving objects with higher efficiency, which avoids the error deduction of some valuable localization references. The fusion method by the NeuroIV and LiDAR features' evaluation method owns smaller localization fluctuation and higher localization accuracy. The outdoors' localization result further verifies the superiority of the feature evaluation method in the more complicated and changeable environment.

VI. Conclusion

In this article, the LiDAR and NeuroIV sensors are combined to get autonomous vehicle's accurate localization in dynamic scenes. Based on the NeuroIV camera's event points, the noise events are filtered with the designed Otsu method, and a new database is presented with a wide variety of NeuroIV categories. Due to the fact that NeuroIV only displays partial body features of the moving objects, a new YOLO-small network is designed to train and obtain the distribution zone of the object's whole body. Through the objects' similarity distribution and matching model, the moving objects' prediction in the NeuroIV camera can be deducted from the LiDAR's whole points, and a remained static environment is acquired.

Due to the sparseness of the LiDAR points, there is a distribution deviation between the extracted feature and the actual feature. Therefore, the uniform distribution and Gaussian distribution models of points' features are established to provide an effective evaluation method. After that, the feature's information and error entropies are solved, which are used to evaluate the features' reliability and optimize the matching weights. According to the evaluated weights, the localization accuracy can be improved greatly.

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